

9708/33/o/n/17

1-C:

A – Museums are not a public good. This because they exhibit the characteristics of **rivalry** as a visit to the museum by an individual may limit how many other people can visit that museum at that particular time because it may get overcrowded. Additionally, it is also **excludable** because all those who do not pay the entrance fee may be excluded from visiting the museum.

B – If social benefits were less than social costs, this represents that museums have a net social cost, and so, the government may not subsidize the cost of entrance.

C – Since social benefits exceed private benefits, this represents the existence of external benefits. This will encourage the government to subsidize the entrance fee so that the society can benefit as a whole. This is why, this becomes correct.

D – If social costs exceed private costs, this represents the existence of external costs.

2-D:

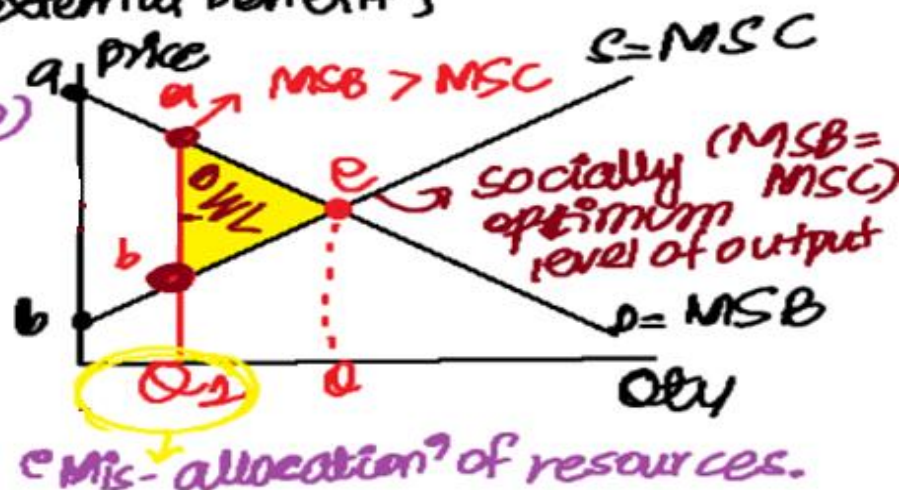
The table shows the marginal costs of production and marginal benefits of consumption.

MEC 50	marginal private cost (\$)	marginal social cost (\$)	marginal private benefit (\$)	marginal social benefit (\$)	MEB 50
	100	150	150	200	

*MEC (marginal external cost)

*MEB (marginal external benefit)

*MSB > MSC
(200) (150)



A and B: Since Marginal external benefit (50) is equal to Marginal external cost(50), we won't say that there is an existence of "a negative consumption externality"(option A) or "a positive production externality" (option B)

C : **Pareto optimality**, also known as **Pareto efficiency**, is a concept in economics that refers to a state where **no individual or group can be made better off without making someone else worse off**. In other words, a situation is considered **Pareto optimal** when **it is not possible to reallocate resources or change conditions to improve the well-being of at least one person without reducing the well-being of another person**.

A **Pareto improvement** occurs when a change makes at least one person better off without making anyone else worse off. **A Pareto optimal outcome, therefore, represents a situation where there are no possible Pareto improvements left.**

This is not illustrated through the table above.

D: Since $MSB > MSC$, it can be said that “under-allocation”(mainly of “merit goods”) is taking place, which results in “market failure” , or in other words, this represents **mis-allocation of resources**. **This can be seen in the diagram above**. As a result of which, key D, becomes correct.

3-C:

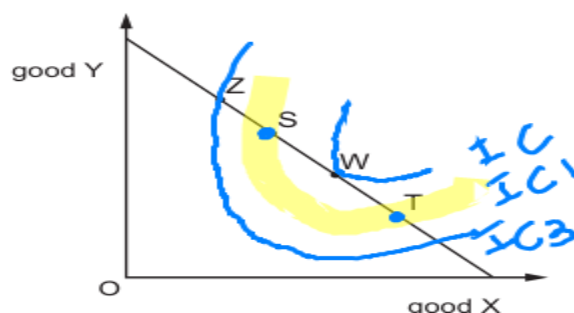
A – This is incorrect, because “**cost benefit analysis**” has a method of valuing the “**long-term benefit**” of the project.

B – This is incorrect, because “**cost-benefit analysis**” is also suitable for **public sector projects** since **it can also take externalities into consideration**. Although, externalities have no market prices attached to them since they do not go through the market , **shadow prices** are attached to them, and then they are included in the “cost-benefit analysis”

C – This is correct because the government may choose to not undertake the project even though it is socially beneficial($SB > SC$), because the project has a “**very high opportunity cost**”, since it requires a waiting period of 10 years. This means that this is a long-term investment , which will yield benefits in the longer term, thus leading to a greater opportunity cost of finances for a longer period of time.

D: There is no reference to the government policy **restricting the extent of the market system**.

4-C:



A – The consumer is not indifferent between combinations W, S and T because they do not lie on the same indifference curve.

B – The consumer does not prefer combination T (which is on the lower indifference curve IC1) to W (which is on the higher indifference curve IC)

C – The consumer will prefer combination W because it is on the higher indifference curve “IC” as compared to combination S (which is on the lower indifference curve “IC1”). This is why, this becomes correct.

D – Again, the consumer does not prefer combination Z to combination T because IC3 gives lesser satisfaction than IC2.

5-A:

A – This is correct because in perfect competition, homogeneous or identical products are produced. Similarly, in monopolistic competition, there is freedom of entry and exit because of less barriers.

B – In perfect competition, there are no barriers to entry (there is costless entry), which makes this option incorrect, although there are differentiated products in monopolistic competition.

C – In perfect competition, there are a very large number of firms as 1000s of producers are engaged in producing standardized products. Whereas, in monopolistic competition, there are not a small number of firms, as there are about 20-25 small firms producing differentiated products.

D – In perfect competition, there is no product differentiation. And also, in monopolistic competition, there are very low barriers to entry.

6-B:

A and C – These are general features of a monopoly, which are not specifically attributed to natural monopolies.

B – Since fixed costs form the greatest proportion of the total costs as with electricity companies, the Average fixed costs keep on falling over a large volume of output, and so, the Average costs also keep on falling too over the entire range of output until the MES is achieved. This can be seen in the diagram below.



D – Dis-economies of scale represent a situation when there is a rise in long-run Average costs as a result of growing beyond the optimal size. In case of a natural monopoly, the Average costs keep on falling over the entire range of output, which represents that dis-economies of scale do not set in.

7-B:

A: Since the firm is already a monopoly producer, it does not need to engage in cooperation or collusion. Such a behavior is usually seen in an oligopoly market structure.

B – Limit pricing is a strategy used by a firm in which it charges a price below its normal profit-maximizing price but above the competitive price line where normal profits are made, **such that the price it charges is below the estimated Average total costs of potential rivals. This is aimed to prevent new firms from entering the market because now they may face the risk of making unsustainable losses. This is why, this becomes correct.**

C – Predatory pricing is a pricing strategy in which a firm charges prices too low, even if it enters a loss-making situation, to **drive existing competitors** out of the market.

D - Price leadership is a situation where a single firm sets the prices for a particular product or service, and other firms in the market follow their pricing decisions. The price leader is usually the dominant firm or the one with the largest market share, and its pricing decisions are considered a benchmark for the industry.

8-C:

A) Barriers to the entry of new firms: This option actually makes collusion easier, not more difficult. With barriers to entry, firms in the industry are protected from new competitors, which reduces the threat of disruption and makes it easier for them to coordinate their actions and maintain collusion.

B) Limited scope for cost reduction via innovation: This option also makes collusion easier, not more difficult. If firms have limited opportunities to reduce costs through innovation, they may be more motivated to collude in order to maintain prices and profits.

C) Significant product differentiation between firms: This option makes collusion more difficult. When firms have significantly different products, they cater to different customer segments and have less incentive to collude. Differentiation also creates competition on attributes other than price, making collusion less effective.

D) Stability in the market demand for the product: This option makes collusion easier, not more difficult. Stable demand reduces uncertainty and makes it easier for firms to coordinate their actions and maintain collusion. Unstable demand, on the other hand, creates uncertainty and makes collusion more difficult.

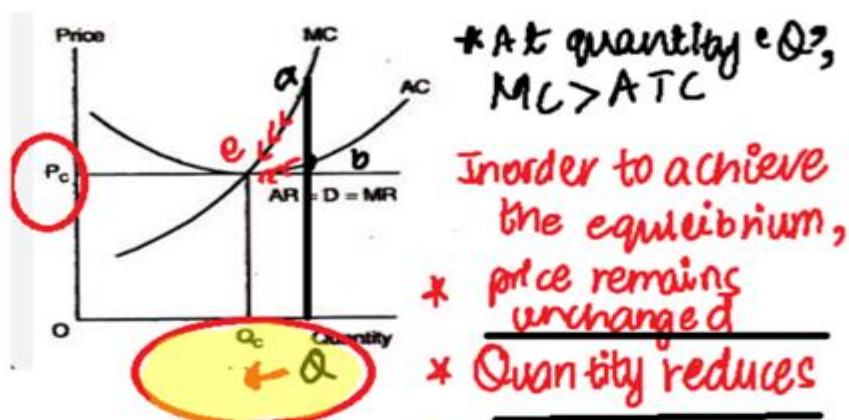
9-C: A – Contestability is a situation in which a potential threat of competition leads to firms behaving as if there was actual competition. This situation arises when there are weak barriers to entry. This is not a defining characteristic of an oligopoly industry.

B – In an oligopolistic industry, there are a small number of competing players, which are large in size.

C - Interdependence in an oligopoly refers to the situation where a small number of firms compete with each other, and **each firm's decision has a significant impact on the others**. This means that each firm's market share is substantial, and changes in one firm's strategy, such as price cuts, can affect the others. As a result, firms in an oligopoly are constantly monitoring and responding to each other's actions, as they are aware of their interdependence. This is a key characteristic of oligopoly.

D – Price-rigidity is a behavior that is usually seen in an oligopolistic industry, that is when prices remain unchanged at the kink, and is shown through the kinked demand curve. This is not a characteristic of an oligopoly. Instead, the characteristic that leads to this behavior is “mutual interdependence”

10-B:



→ In order to achieve the equilibrium (e) in the long-run, the firm must operate on the long-run equilibrium where $MC = MR$

→ When the price remains unchanged, and the quantity (output) decreases from Q to Q_1 , the equilibrium is achieved.

This is why, key B, becomes correct.

11-B:

A and C: Banking and insurance are service sector firms, and they are not in the industrial sector.

B: Small firms can easily take up small construction projects that require lesser capital investment, making it easier for them to survive.

D- Small firms may struggle to survive in the mining industry due to various challenges. One major obstacle is the high capital requirement, as mining demands significant investment in equipment, infrastructure, and technology. Additionally, large firms can reduce costs per unit by mining on a larger scale, making it difficult for small firms to compete.

12-A:

A) A **contestable market** is a market where entry and exit barriers are low, and firms can easily enter or exit the market. To avoid attracting competition from new entrants, a firm in a contestable market will charge the maximum price (**or in other words it will employ limit pricing strategies**) that is consistent with this aim, as any higher price would attract new entrants.

B) Monopolistically competitive market: In this market, firms have some degree of market power, and the potential threat of competition may be limited if there are some barriers arising from existing firms engaging themselves in product differentiation.

C) Monopsonistic market: This market refers to a labor market with a single buyer of labor, and is not relevant to the question.

D) Perfectly competitive market: In this market, there are many firms, free entry and exit, and perfect information. A firm in a perfectly competitive market would not be able to charge a price to avoid competition. This is because, a single producer has no influence over the market price and has to accept the price set in the market.

13-D:

A - In a **monopolistically competitive market**, a change in the output of one firm can affect the industry price. This is because firms in this market structure have **some degree of market power**. While there are many firms producing similar products, each firm has a degree of product differentiation, **which means that a firm may have a higher demand than the other in the short-run**, making it able to charge higher prices. As a result, a change in output by one firm can lead to a change in the industry price.

B - In a monopoly, the single firm has complete market power, and a change in its output will directly affect the industry price. The monopolist can set prices and output levels to maximize profits, and its output decisions are not influenced by other firms. As a result, a change in output by the monopolist will lead to a change in the industry price.

C - In an oligopolistic market, a change in output by one firm can affect the industry price. **This is because firms in this market structure interact strategically, and their output decisions are influenced by the expected reactions of their competitors.** With only a few firms producing similar products, each firm has a degree of market power, and its output decisions can influence the market supply curve. As a result, a change in output by one firm can lead to a change in the industry price.

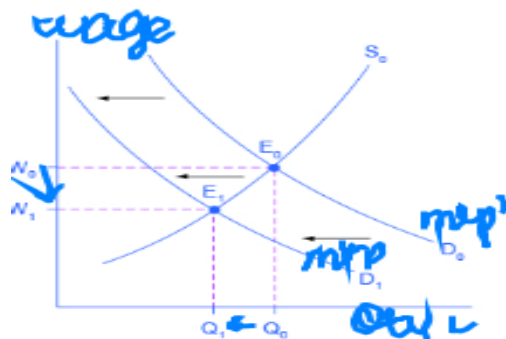
D: In a perfectly competitive market, the industry price remains unaffected by a change in the output of one individual firm. This is because each firm has an extremely small market share compared to the overall market demand, and so a single firm's actions would not affect the market price.

14-C:

A – This is incorrect because there would not be a movement down the marginal revenue product curve, instead the MRP curve will shift leftwards.

B and D – Shift towards the horizontal axis of the Marginal cost curve(option B) and shift towards the vertical axis of the average cost curve(option D), refers to the product market and not labor market.

C: As a result of a fall in the demand of oil, when the price of oil reduces, this will result in a fall in the **derived demand for workers in the oil industry**, which will result in a rise in unemployment, represented through a loss in 65000 jobs. Since the demand for labor is also known as the “marginal revenue product” curve, it will result in a shift towards the left or towards the origin of the Marginal revenue product curve. As a result of which, key C, becomes correct.



15-A:

Transfer earnings represent the minimum wage required to keep a worker in the industry, or in other words, it is regarded as **the opportunity cost of keeping a labor in its present use, and since it is determined by what the labor would want in its next best employment**, the \$60000 that would have been earned as a lecturer represents “transfer earnings”

And, so the economic rent will be \$40,000(100,000-60,000), which is the amount over and above the transfer earnings.

This explains why, key A, will be correct.

16-D:

A – When someone has been given property rights, this means that they own that particular property, and so, **private owners will be encouraged to conserve their property when they expect that the value will rise in future.**

B - When someone has been given property rights, this means that they own that particular property, and so, they will be incentivized to **manage their property carefully.**

C – If I own a particular piece of land, on which I have built a factory, which leads to “pollution”, because of these property rights, I will be accountable, or in other words, bound by “law”, to compensate for the damage I’ve caused to the people living nearby, as a result of “**property misuse**”

D – Existence of property rights do not prevent private owners gaining from using resources to benefit others. Since, this is **not a possible outcome, this option becomes correct.**

17-A:

A - When a state-owned industry is privatized as a monopoly, the **absence of barriers to entry increases the likelihood of achieving productive efficiency.** This is because potential competitors can enter the market, creating a threat of competition that incentivizes the monopolist to operate efficiently. This is why, this becomes correct.

B- High sunk costs, such as significant capital requirements, can actually create a barrier to entry for potential competitors. This reinforces the monopolist's market power, making it more difficult for new firms to enter the market. As a result, the monopolist may feel less pressure to achieve productive efficiency.

C - Significant economies of scale can reinforce the monopolist's market power, making it more difficult for new firms to compete. When a firm has economies of scale, it can produce at a lower cost per unit than its competitors, making it harder for new entrants to compete.

D - If variable costs form a small proportion of total cost, this represents that fixed costs form a greater proportion of total costs. This further means that there is a potential to gain significant benefits from economies of scale for these firms as the fixed costs keep on falling over a large volume of output. Significant economies of scale can then reinforce the monopolist's market power, making it more difficult for new firms to compete.

18-D:

A – A switch on taxation from income tax to taxes on expenditure represents that the taxation system has been switched from progressive to regressive. Since in case of regressive taxes, the average rate of tax falls upon a rise in income, and the marginal rate of tax is lesser than the average rate of tax, which represents that for every extra \$1 earned, less is taken up in tax, this indicates that they take up a greater proportion of incomes of the poor than the rich. This is why, they may result in inequitable outcomes.

B – If by means of minimum price legislation, the prices of essential foodstuffs are raised, this will increase total expenditure of the low income groups, since the quantity demanded of these goods cannot be contracted by much as they have an inelastic demand. This is why, they may result in inequitable outcomes.

C – The privatization of state run transport systems will also lead to inequitable outcomes as in pursuit of maximizing profits, the private firm is most likely to charge higher prices for transport.

D - When people have access to free healthcare, they are more likely to seek preventive care, manage chronic conditions, and receive timely treatment for illnesses, resulting in a **healthier and more productive workforce**. This, in turn, leads to increased economic output, as a healthy workforce is more efficient and productive. The supply of free healthcare can improve **equity** by reducing health disparities and increasing access to healthcare for vulnerable populations by ensuring that everyone has equal access to necessary care.

19-D:

Adult literacy rate (A) and average life expectancy at birth (B) are both indicators of human development and well-being, which are closely related to economic development. Level of real GNP per head (C) is a direct measure of economic performance and development.

Size of population (D), on the other hand, is not a direct indicator of economic development. A large population does not necessarily mean a country is more economically developed, as it depends on various factors like productivity, income distribution, and resource utilization.

Therefore, size of population is the least likely indicator to be used when measuring comparative economic development

20-D:

Nominal GDP can be measured through:

$$C+I+G+(X-M)$$

It can be seen above that nominal GDP already accounts for the effects of exports and imports, which is why, option A, B and C become incorrect.

By adjusting nominal GDP figures for inflation and the population change, we can obtain data of real GDP per head, which is a relatively better measure of comparative living standards. This is why, option D becomes correct

21-D:

A – Although the percentage of population of the school going age from 4-18 is lesser in 2050 than in 2015, it can still not be said that fewer school places will be needed in 2050. This is because, it might be that even though the percentage of school going children is less, the overall size of population may have been higher in 2050, which means that there may be more children in absolute terms in that year.

B – Based on data on population structure alone, we cannot deduce the level of employment in any year.

C - Based on data on population structure alone, we cannot deduce the number of people employed between the ages 20-24 in any year.

D – In 2050, there is a relatively higher percentage of population belonging to the retirement age i.e. above 65 years, as compared to 2015. This is why, this becomes correct.

22-C:

- 22 The table gives information on differences between females and males in 2013 in the Human Development Index and Gross National Income per capita in countries with different levels of development.

	HDI		GNI per capita (US \$ purchasing power parity 2011)	
	female	male	female	male
very high development	.871	.887	26 667	53 681
high development	.710	.733	9 426	16 966
medium development	.565	.622	3 199	8 619
low development	.446	.506	2 011	3 789

Handwritten notes:

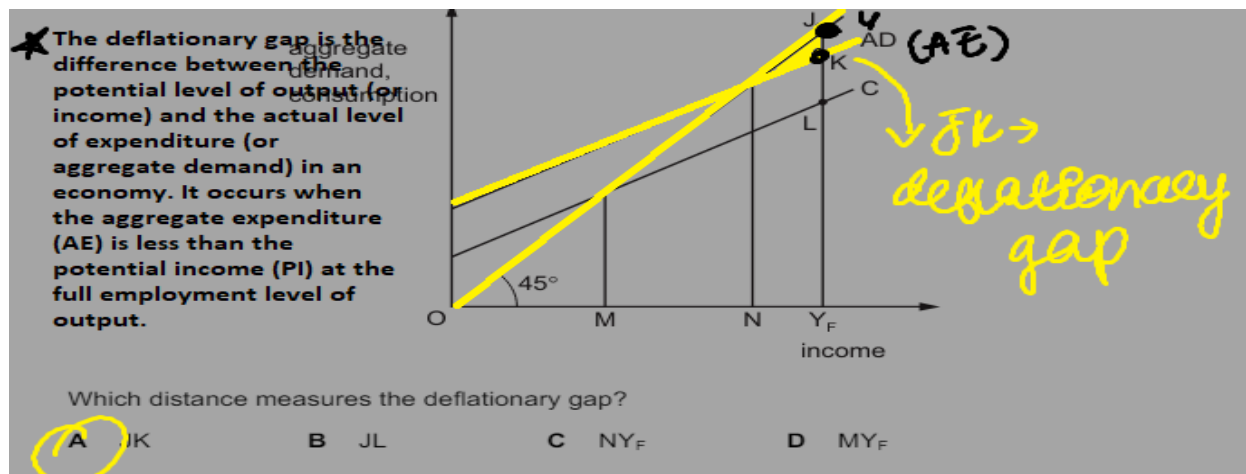
- Blue:* Difference b/w males & female HDI.
 - 16 ← very high development
 - 23 ← high development
 - 57 ← medium development
 - 60 ← low development
- Pink:* Difference b/w males & female gni/cap
 - 27 014 ← very high development
 - 7 540 ← high development
 - 5 420 ← medium development
 - 1 778 ← low development

- A – This option is incorrect because females are closer to males in their **development opportunities** as compared to their **earning abilities**.
- B – This is incorrect because females are closest to males in development opportunities in countries with very high income development.
- C – In countries with very high income development, males have a gni per capita of 53681 which is 2.013 times higher than that of females who have that of 26667. This means that it will be correct to say that **males have over twice the purchasing power of females in countries with very high development**.
- D – This table includes data of HDI

23-B:

- $Y = C + I + G + (X - M)$
- $14 = 40 + 0.54 + 80 + (100 - 100)$
- $14 = 120 + 0.54$
- $14 - 0.54 = 120$
- $0.54 = 120, \quad Y = \frac{120}{0.5} \rightarrow 240$

24-A:



It is to be noted that N-Y_f (key C) represents the negative output gap since it shows the difference between the actual income and the potential income.

25-A:

Velocity of circulation in the quantity theory of money refers to the number of times money changes hands within a certain period.

A – A move to monthly salary payments would tend to reduce the value of V, the income velocity of circulation. This is because individuals would hold onto their money for longer periods, reducing the frequency of transactions and thus the velocity of money. With weekly payments, individuals are more likely to spend their money soon after receiving it, increasing the velocity of money. By contrast, monthly payments would lead to a longer average holding period, reducing V.

B – As interest rates rise, the velocity of money increases. Also, it is logical, because as the interest rate rises, rather than holding money people will try gaining advantages from the high interest rate. Hence, the reduction in average holding of money causes the velocity of money to rise.

C – An increase in the use of cash machines means that people can withdraw more to spend which ultimately increases spending, and this rise in transactions will result in an increase in frequency of money circulation. This can have the effect of an increase in “V”

D - An increase in credit card usage would also increase the velocity of money, not reduce it. Credit cards facilitate transactions and increase the speed of money circulation.

26-A:

Circular Flow of Income

In the circular flow of income, households provide factor services (such as labor, land, and capital) to firms in exchange for income (wages, rent, and interest).

27-D:

Liquidity Preference

Liquidity preference refers to the preference of individuals and businesses to hold their wealth in the form of money rather than in other assets.

A) A desire to be paid in cash rather than by cheque: This is a preference for payment method, not liquidity preference.

B) A desire to be paid monthly rather than weekly: This is a preference for payment frequency, not liquidity preference.

C) A desire to hold assets that are convertible into cash: liquidity preference specifically refers to holding money itself, not assets that can be converted into money.

28-C:

Deflationary Gap

A deflationary gap occurs when aggregate demand is below the level required to achieve full employment.

Appropriate Policy

To close a deflationary gap, the government should increase aggregate demand. An increase in government spending (C) is an appropriate policy, as it directly injects money into the economy, stimulating Aggregate demand.

A) An increase in the rate of interest: This would actually worsen the deflationary gap, as higher interest rates reduce borrowing and spending, which will dampen Aggregate demand further.

B) An open market sale of bonds: This would also worsen the deflationary gap, as it reduces the money supply and increases interest rates. This will reduce borrowing and spending, which will dampen Aggregate demand further.

D) An increase in income tax: This would reduce disposable income, leading to lower consumption and aggregate demand, worsening the deflationary gap.

29-B:

A **poverty trap** occurs where an individual or household is discouraged from working or increasing their income because the resulting loss of means-tested benefits and also the obligation to pay more tax as income exceeds a certain level, would leave them no better off, or even worse off, than they were before.

Why means-tested benefits can lead to a poverty trap?

1. **Benefits cliff:** As income rises, benefits are withdrawn, leading to a significant loss of income.

This explains why key B (means tested benefits) is correct.

A) Legal minimum wage: This policy aims to protect low-income workers, and it doesn't create a poverty trap. Instead, this will incentivize workers to work.

C) Prevention of cheaper imports: This policy is related to trade and protectionism, not directly related to poverty traps.

D) Proportional taxation: This tax system takes the same percentage of income from all individuals, regardless of income level, and may not create a poverty trap.

30-B:

Note: **Year 1: When the government reduces indirect taxation, the price of goods and services fall, as a result of which, consumption will increase.**

Poor households, with their higher MPC, are likely to decrease their saving in Year 1, as they take advantage of the temporary tax reduction to increase their consumption as now they spend a higher proportion of their additional real income on consumption.

However, an equivalent increase in direct taxation results in a decrease in disposable incomes, which is likely to reduce savings as well.

A lower disposable income represents that less is available to be saved.

Therefore, the overall impact will be a reduction in savings in year 1.

- **Year 2:** The temporary policies imposed in Year 1 are reversed.
- The impact on savings is reversed, leading to an increase in savings in Year 2.

Therefore, the overall impact will be an increase in savings in year 2.